

Area Unit Study Guide

Created By Gabrielle Cox

1 If the radius of a circle is 10 cm, what will be the area of the circle?

Use $\pi = 3.14$

☐ A 314 cm^2

☐ B 78.5 cm^2

☐ C 100 cm^2

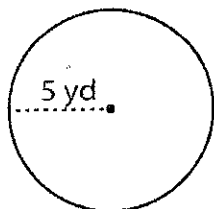
☐ D 31.4 cm^2

$$A = \pi r^2$$

$$A = 3.14 \cdot 10$$



2 Find the area and circumference of the following circles.



a.

$$A = \pi r^2$$

Area:

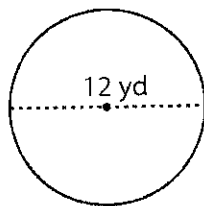
78.5 yd

$$C = 2\pi r$$

$$A (3.14) (5)$$

Circumference:

345.4



b.

$$A = \pi r^2$$

Area:

113.04 yd

$$3.14 \cdot 12^2$$

Circumference:

37.6

a

yd²

b

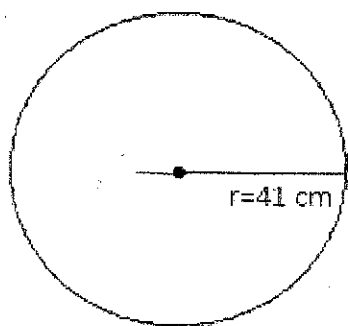
yds

c

yd²

d

yds



$$C = 2\pi r$$

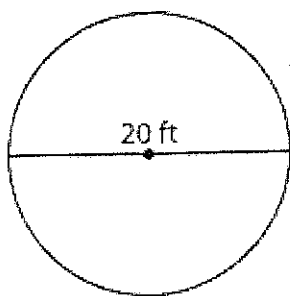
$$C = 2 \cdot 3.14 \cdot 41$$

Answer:

257.48 cm

cm

Find the circumference and the area of the following circle.



The circumference is



The area is



2

 mi^2 ft^2 feet

2

lb

lb

lb

 mi^2

oz

lb

lb

second

in

hour

yard

cup

qt

tbsp

ton

tsp

mph

fps

 ft/sec^2 in^3 mi^3 ft^3

2

 mi^2 ft^2 feet

2

lb

lb

lb

 mi^2

oz

lb

lb

second

in

hour

yard

cup

qt

tbsp

ton

tsp

mph

fps

 ft/sec^2 in^3 mi^3 ft^3

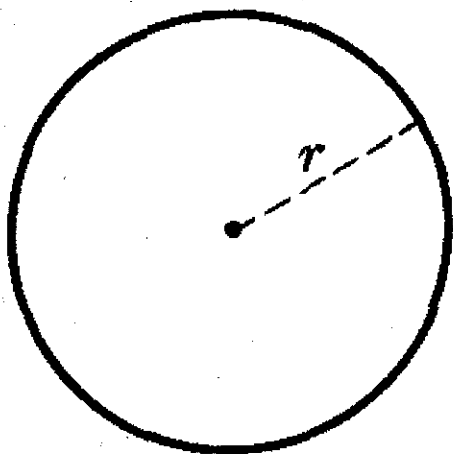
5

The diameter of a circle is 8 centimeters. What is the area?

Area = cm^2

Use $\pi = 3.14$

Use the picture of the circle to choose the correct area formula(s).



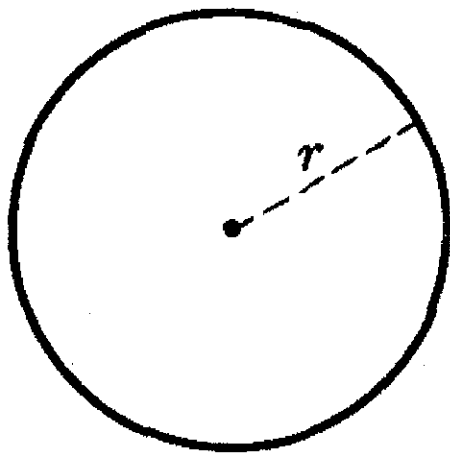
☐ A $A = LW$

☐ B $A = \pi r^2$

☐ C $A = \frac{1}{2}bh$

☐ D $A = \frac{(b_1 + b_2)}{2} \cdot h$

☐ E $A = s^2$



- A** $P = a + b + c$
- B** $P = 2L + 2W$
- C** $P = s_1 + s_2 + s_3$
- D** $P = 2\pi r$
- E** $P = L + W + L + W$
- F** $P = 4s$

The line that runs the circle in half is known as the _____.

The distance from the center of the circle to the edge is known as the _____.

The distance around the edge of the circle is known as the _____.

The space inside the circle in square units is known as the _____.

a

radius
diameter
circumference
area

b

radius
diameter
circumference
area

c

area
radius
circumference
diameter

d

area
radius
circumference
diameter

Find the radius of each circle given the diameter and match them with their appropriate value.

Diameter = 40.67 cm

Diameter = 24.43 cm

Diameter = 18.95 cm

Diameter = 17.34 cm

DRAG & DROP THE ANSWER

25.55 cm

13.33 cm

12.22 cm

8.67 cm

20.34 cm

9.48 cm

Calculate the diameter of a circle for each radius that is given and fit with their appropriate result.

radius = 87.2 cm

radius = 65.9 cm

radius = 72.3 cm

radius = 56.7 cm

DRAG & DROP THE ANSWER

114.6 cm

174.4 cm

131.8 cm

113.4 cm

11 Write the formula for the circumference of a circle.

C =

12 A circle has a radius of 12 centimeters. Which expression gives the circumference of the circle, in centimeters?

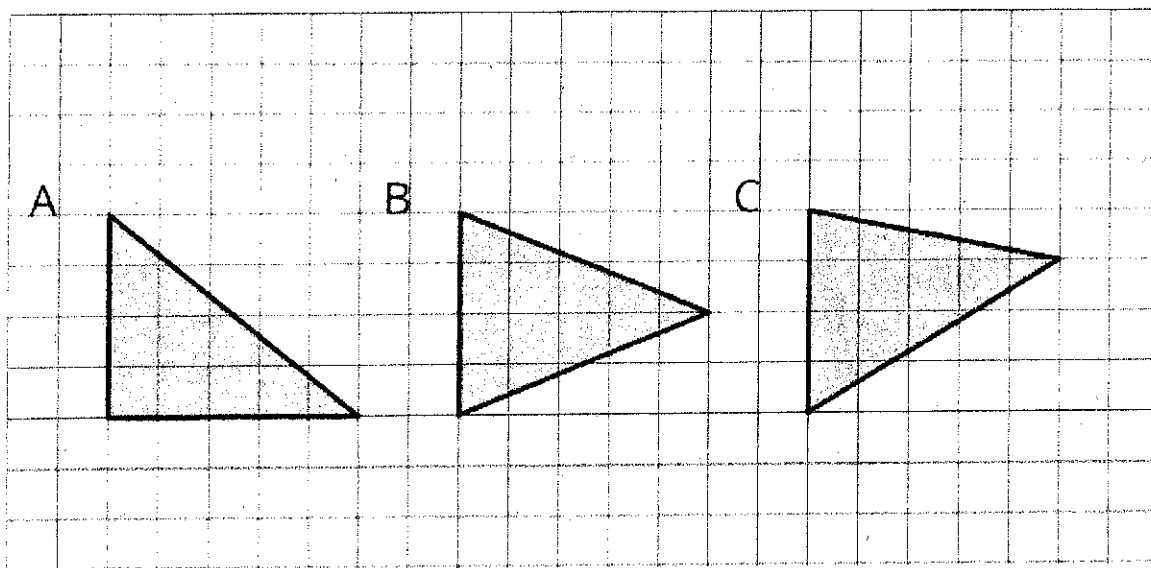
(A) $\pi(12)$

(B) $2\pi(12)$

(C) $\pi(12^2)$

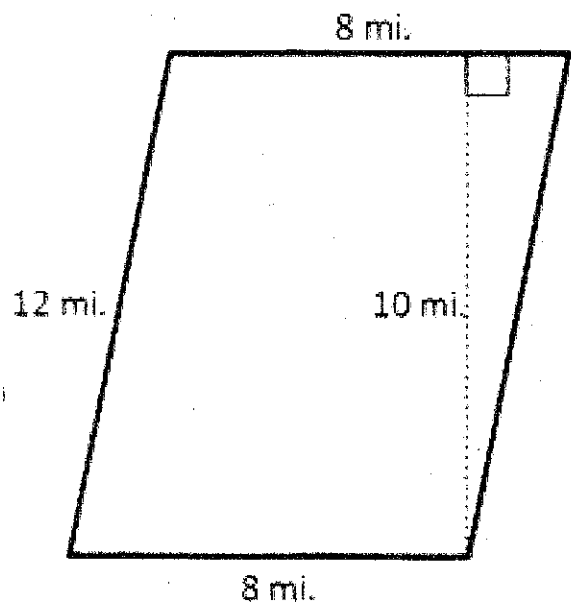
(D) $2\pi(12^2)$

Which of the three triangles has the greatest area? Show your reasoning.



If you get stuck, use what you know about the area of parallelograms to help you.

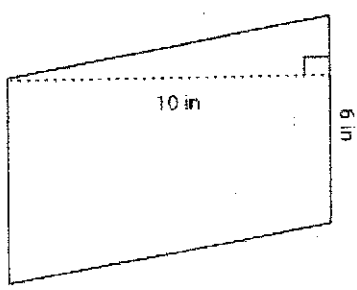
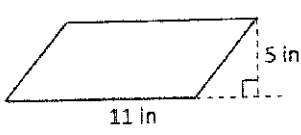
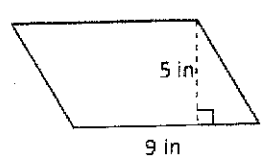
Find the area of the parallelogram below.



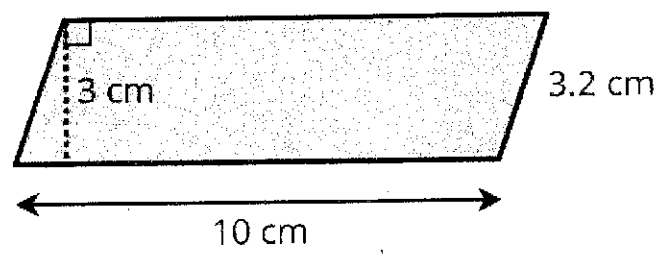
Area of parallelogram =

Find the area of the following parallelograms and choose the appropriate result.

- Parallelogram 55 in^2 45 in^2 60 in^2 30 in^2

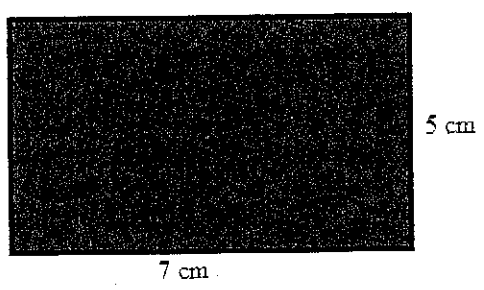


Find the area of the parallelogram.



sq cm

Find the area of the rectangle.



Area = cm^2

2

in^2

ft^2 feet

2

1

1

1

1

oz

1

1

second

in

our

mid

cup

qt

tbsp

ton

tsp

mph

fps

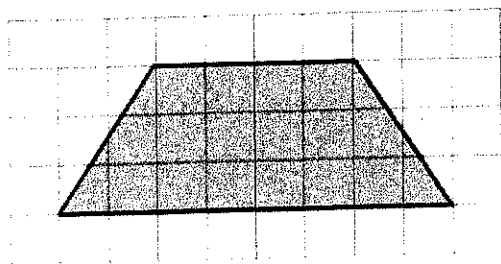
ft/sec^2

in^3

mi^3

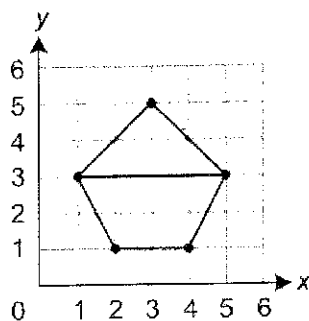
ft^3

Find the area of this trapezoid. Explain or show your strategy.



square units.

A triangle and a trapezoid are graphed on the coordinate plane shown below.



What is the combined area, in square units, of the two shapes?

- (A) 10
- (B) 14
- (C) 16
- (D) 20

Triangle

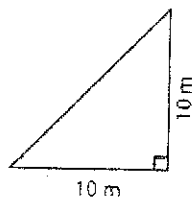
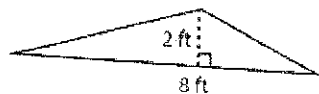
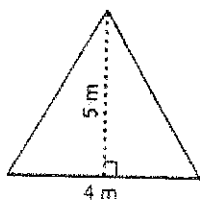
100

8

10

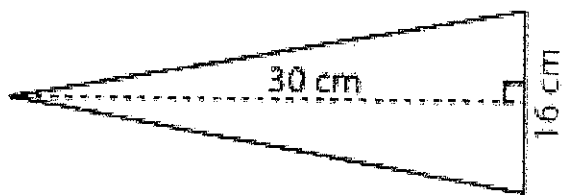
50

20



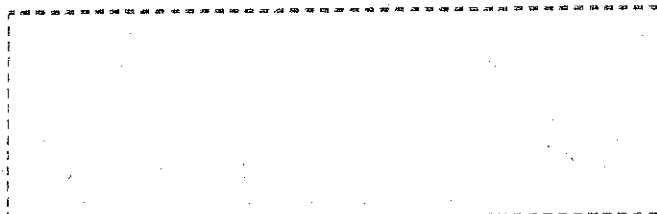
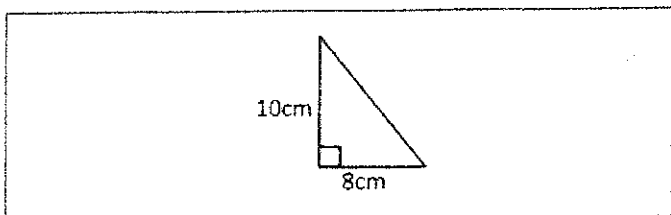
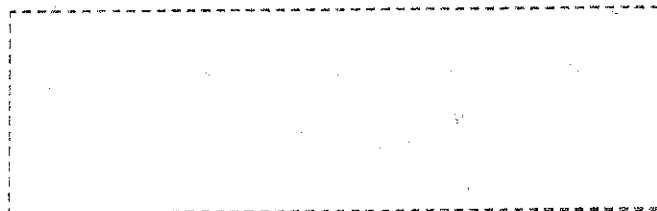
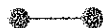
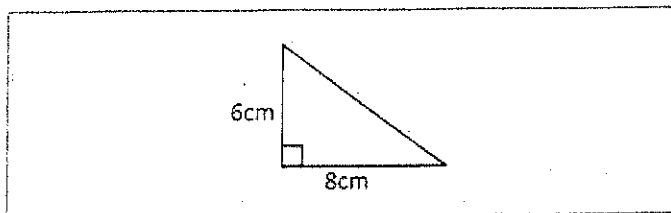
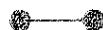
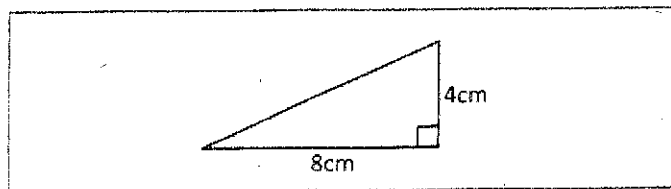
21

Find the area of the triangle and choose the appropriate result.



- (A) 120 cm^2
- (B) 240 cm^2
- (C) 220 cm^2
- (D) 480 cm^2

Find the area of the following triangle and match them with their appropriate result.



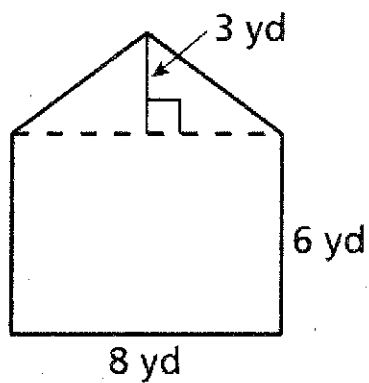
DRAG & DROP THE ANSWER

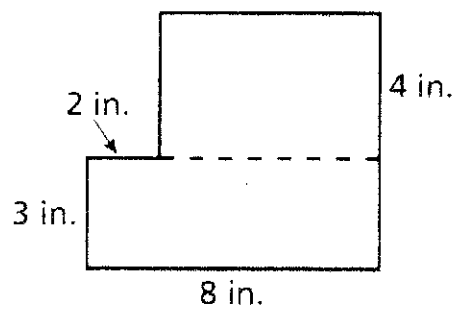
24 cm²

16 cm²

40 cm²

Find the area of the composite figure.





Find the area of the composite figure below.

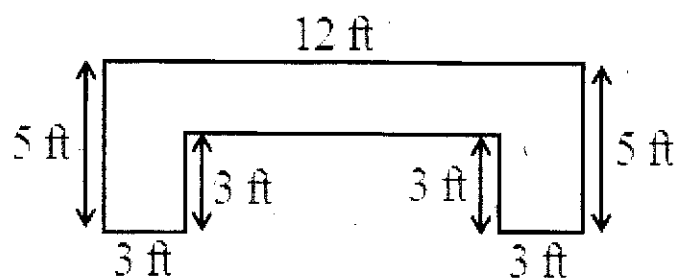
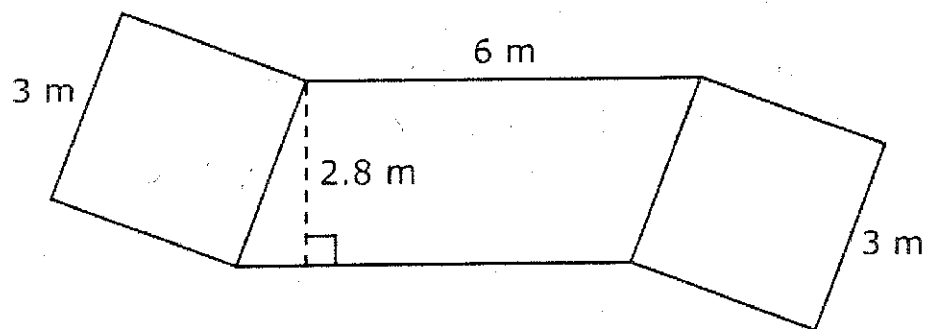


Figure not drawn to scale.

Answer:

ft²

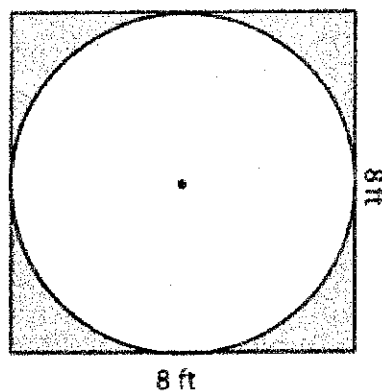
Two congruent squares and a parallelogram were used to form the figure shown.



What is the area of the figure in square meters?

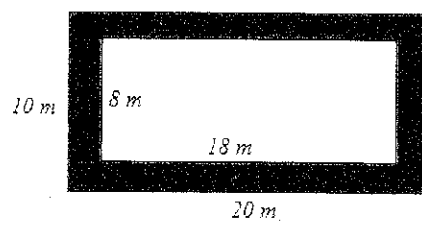
- (A) 25.8 m^2
- (B) 40.8 m^2
- (C) 34.8 m^2
- (D) 28.8 m^2

Find the area of the shaded region. Use 3.14 for π . Round your answer to the nearest hundredth.



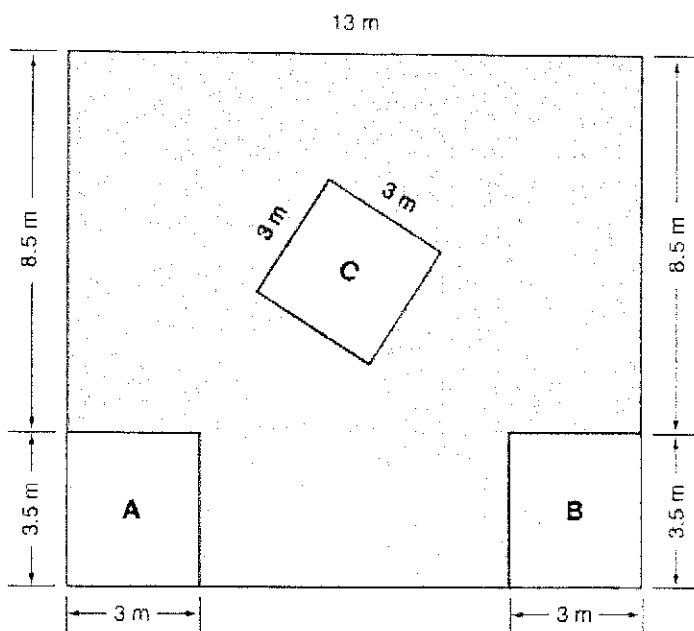
Area of shaded region =

Find the area of the blue shaded region.



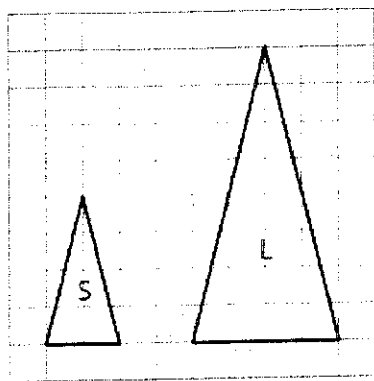
- (A) 144 m^2
- (B) 200 m^2
- (C) 56 m^2
- (D) 344 m^2

Find the area of the shaded portion and type your result in the empty box.



Answer: m^2

Triangle S and Triangle L are scaled copies of one another.

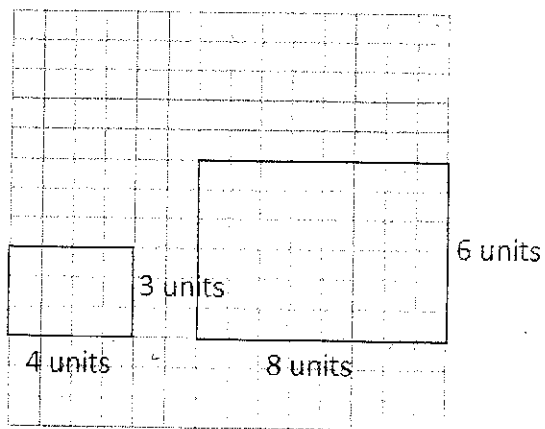


1. What is the scale factor from S to L ?

2. What is the scale factor from L to S ?

3. Triangle M is also a scaled copy of S . The scale factor from S to M is $\frac{3}{2}$. What is the scale factor from M to S ?

What is the scale factor the smaller rectangle to the larger rectangle?



- (A) 6 units
- (B) 18 units
- (C) 2 units
- (D) 48 units

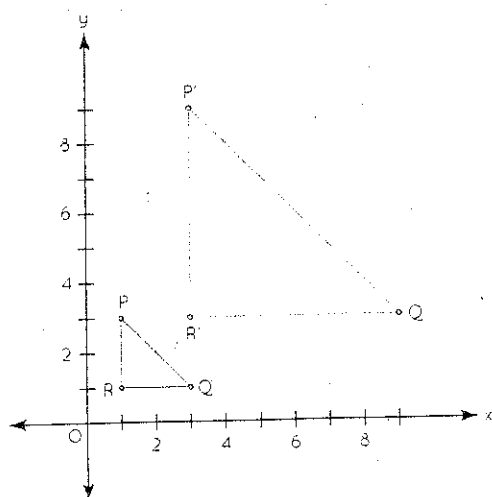
What type of dilation do you have with a scale factor of $\frac{1}{3}$?

- (A) enlargement
- (B) reduction
- (C) none of the above

What type of dilation would you have with a scale factor of 3?

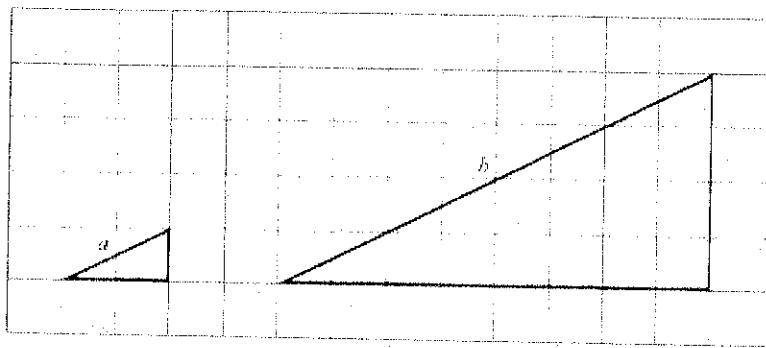
- (A) enlargement
- (B) reduction
- (C) none of the above

What is the scale factor from the smaller triangle to the larger triangle?



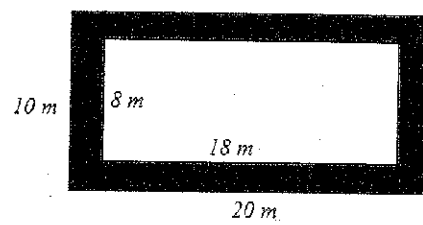
- (A) 6
- (B) 2
- (C) 3
- (D) 4

g. Which figure shows triangle a dilated to triangle b?



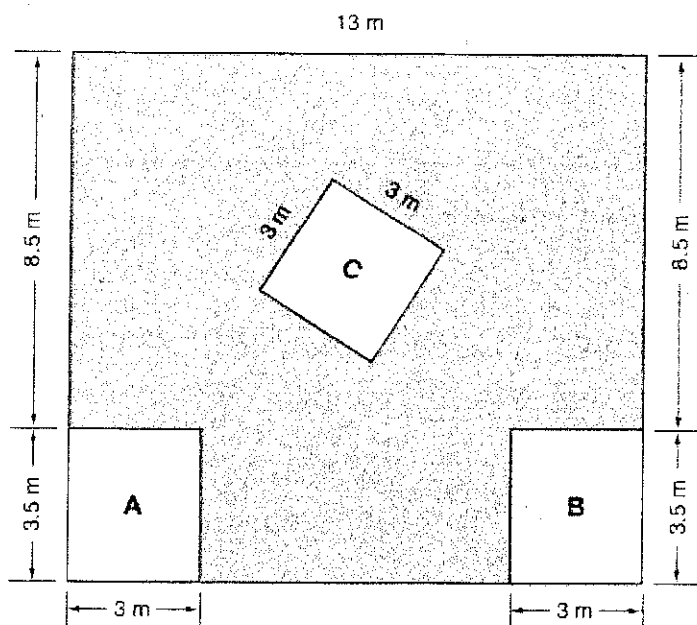
- (A) $\frac{8}{2} = \frac{1}{1} = 4$
- (B) $\frac{2}{8} = \frac{1}{4} = 0.25$
- (C) $\frac{6}{3} = \frac{2}{1} = 2$
- (D) $\frac{3}{6} = \frac{1}{2} = 0.5$

Find the area of the blue shaded region.



- (A) 144 m^2
- (B) 200 m^2
- (C) 56 m^2
- (D) 344 m^2

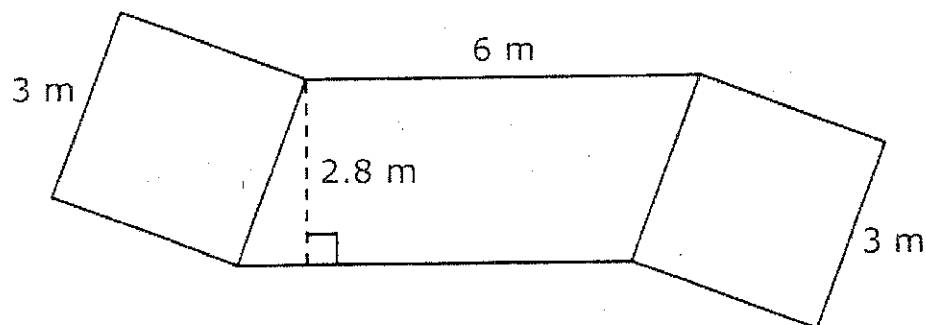
Find the area of the shaded portion and type your result in the empty box.



Answer:

m^2

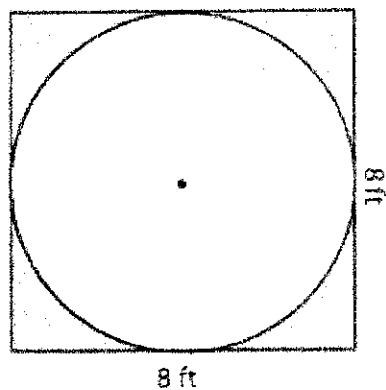
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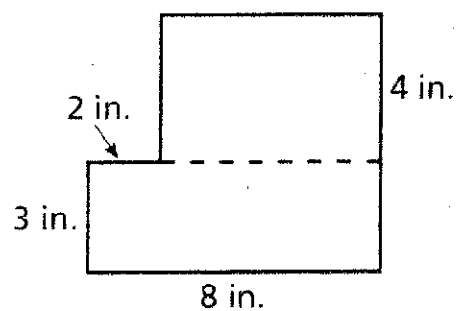
What is the area of the figure in square meters?

- (A) 25.8 m^2
- (B) 40.8 m^2
- (C) 34.8 m^2
- (D) 28.8 m^2

Find the area of the shaded region. Use 3.14 for π . Round your answer to the nearest hundredth.



Area of shaded region =



Find the area of the composite figure below.

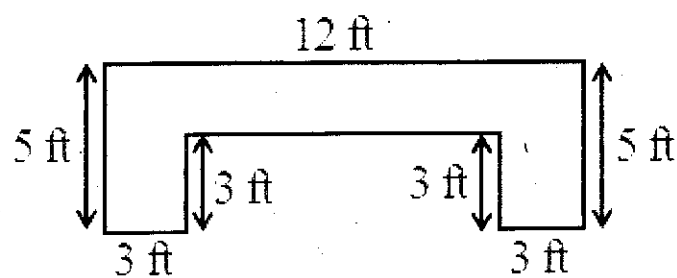
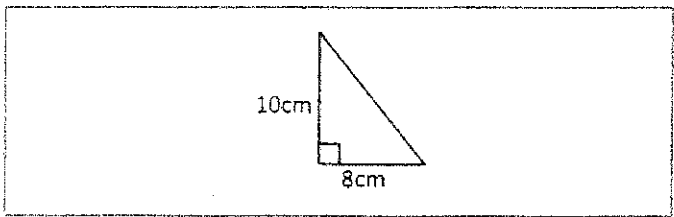
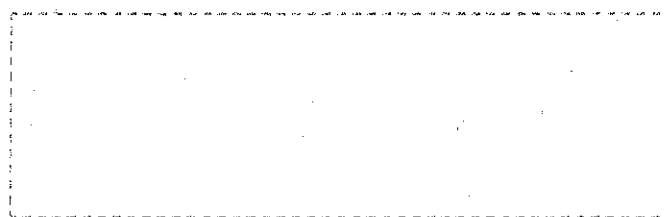
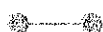
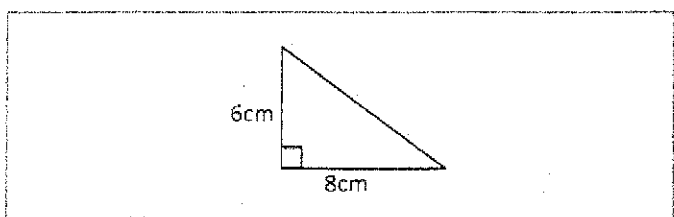
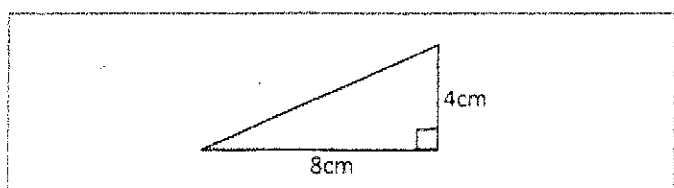


Figure not drawn to scale.

Answer: ft²

Find the area of the following triangle and match them with their appropriate result.



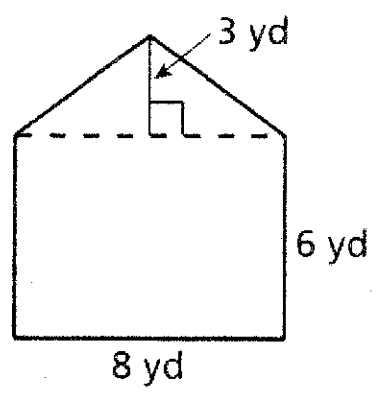
DRAG & DROP THE ANSWER

24 cm²

16 cm²

10 cm²

Find the area of the composite figure.



Find the area of each triangle and choose the appropriate result.

Triangle

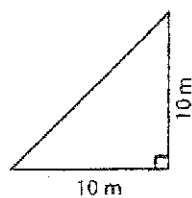
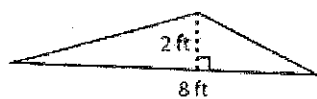
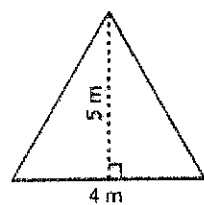
100

8

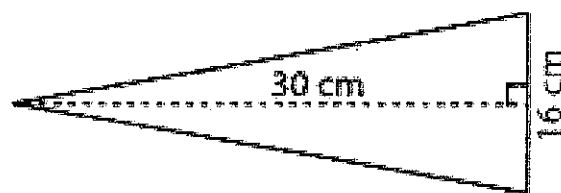
10

50

20

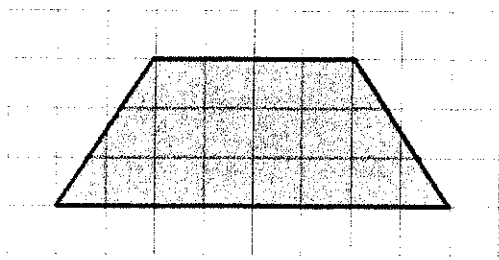


Find the area of the triangle and choose the appropriate result.



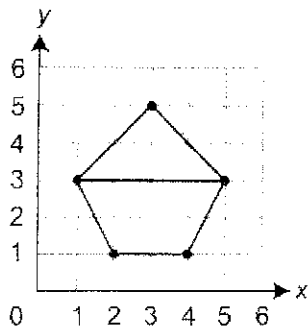
- (A) 120 cm^2
- (B) 240 cm^2
- (C) 220 cm^2
- (D) 480 cm^2

18 Find the area of this trapezoid. Explain or show your strategy.



square units

19 A triangle and a trapezoid are graphed on the coordinate plane shown below.



What is the combined area, in square units, of the two shapes?

- (A) 10
- (B) 14
- (C) 16
- (D) 20

2

in²

ft² feet

1

lb

in

oz

in

oz

lb

in

second

in

our

mi

cup

qt

tbsp

ton

tsp

mph

fps

ft/sec²

in³

mi³

ft³

Find the area of the following parallelograms and choose the appropriate result.

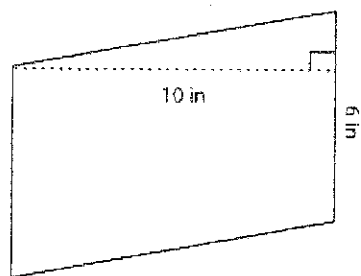
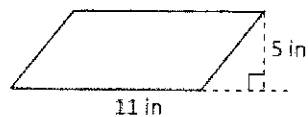
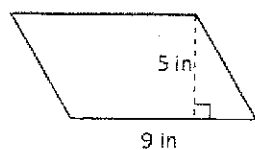
Parallelogram

55 in^2

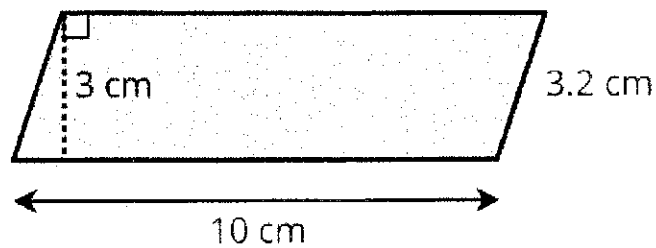
15 in^2

60 in^2

30 in^2

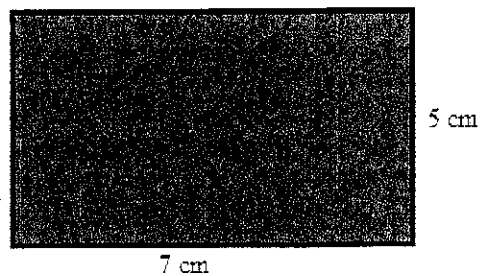


16 Find the area of the parallelogram.



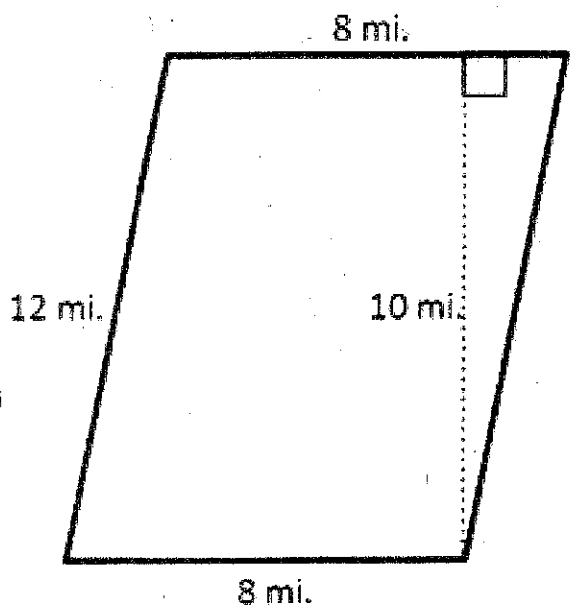
sq cm

17 Find the area of the rectangle



Area = cm^2

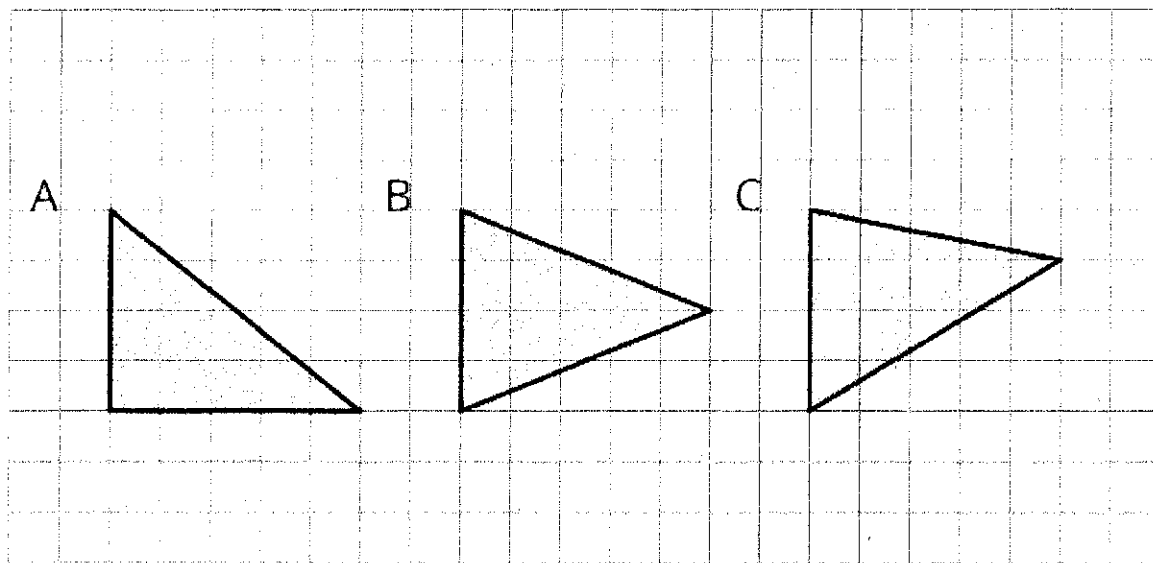
Find the area of the parallelogram below.



Area of parallelogram =



Which of the three triangles has the greatest area? Show your reasoning.



If you get stuck, use what you know about the area of parallelograms to help you.

Calculate the diameter of a circle for each radius that is given and it with their appropriate result.

radius = 87.2 cm



radius = 65.9 cm



radius = 72.3 cm



radius = 56.7 cm



DRAG & DROP THE ANSWER

144.6 cm

174.4 cm

131.8 cm

113.4 cm

Write the formula for the circumference of a circle.

C =

A circle has a radius of 12 centimeters. Which expression gives the circumference of the circle, in centimeters?

(A) $\pi (12)$

(B) $2\pi (12)$

(C) $\pi (12^2)$

(D) $2\pi (12^2)$

Find the radius of each circle given the diameter and match them with their appropriate value.

Diameter = 40.67 cm

Diameter = 24.43 cm

Diameter = 18.95 cm

Diameter = 17.34 cm

DRAG & DROP THE ANSWER

25.55 cm

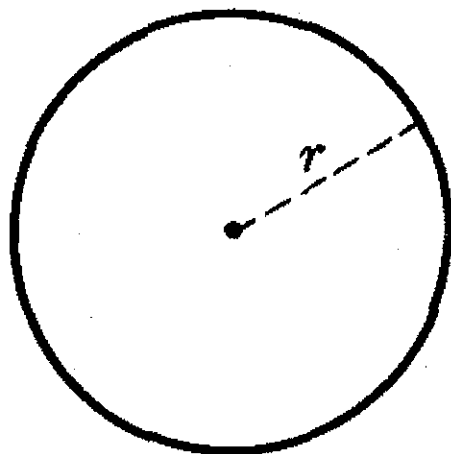
13.33 cm

12.22 cm

8.67 cm

20.34 cm

9.48 cm



- A** $P = a + b + c$
- B** $P = 2L + 2W$
- C** $P = s_1 + s_2 + s_3$
- D** $P = 2\pi r$
- E** $P = L + W + L + W$
- F** $P = 4s$

8

The line that cuts the circle in half is known as the **a**

The distance from the center of the circle to the edge is known as the **b**

The distance around the edge of the circle is known as the **c**

The space inside the circle in square units is known as the **d**

a

radius

diameter

circumference

area

b

radius

diameter

circumference

area

c

area

radius

circumference

diameter

d

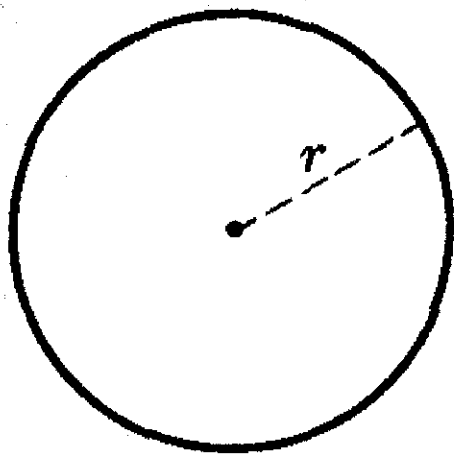
area

radius

circumference

diameter

Use the picture of the circle to choose the correct area formula(s).



☐ A $A = LW$

☐ B $A = \pi r^2$

☐ C $A = \frac{1}{2}bh$

☐ D $A = \frac{(b_1 + b_2)}{2} \cdot h$

☐ E $A = s^2$

2

in^2

ft^2 feet

2

in^2

ft^2

in^2

in^2

oz

in^2

in^2

second

in^2

hour

yard

cup

qt

tbsp

ton

tsp

mph

fps

ft/sec^2

in^3

mi^3

ft^3

2

in^2

ft^2 feet

2

in^2

ft^2

in^2

in^2

oz

in^2

in^2

second

in^2

hour

yard

cup

qt

tbsp

ton

tsp

mph

fps

ft/sec^2

in^3

mi^3

ft^3

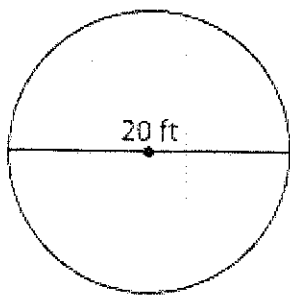
5

The diameter of a circle is 3 centimeters. What is the area?

Area = cm^2

Use $\pi = 3.14$

Find the circumference and the area of the following circle.



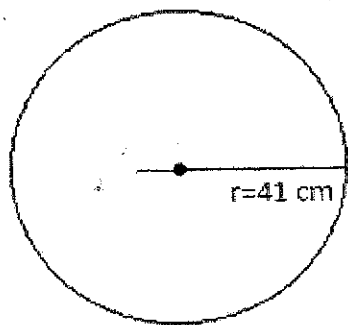
The circumference is:



The area is:



Calculate circumference of the circle. (use $\pi = 3.14$)



$$C = 2\pi r$$
$$C = 2 \cdot 3.14 \cdot 41$$

Answer:

257.48 cm

cm

Area Unit Study Guide

Created By Gabrielle Cox

- 1 If the radius of a circle is 10 cm, what will be the area of the circle?

Use $\pi = 3.14$

☐ A 314 cm²

☐ B 78.5 cm²

☐ C 100 cm²

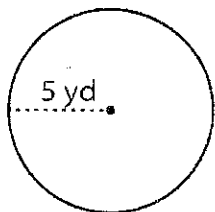
☐ D 31.1 cm²

$$A = \pi r^2$$

$$A = 3.14 \cdot 10$$



- 2 Find the area and circumference of the following circles.



a.

$$A = \pi r^2$$

Area:

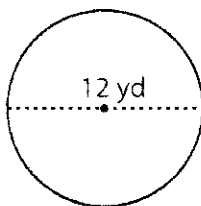
78.5 yd

$$C = 2\pi r$$

$$A (3.14) (5^2)$$

Circumference:

345.4



b.

$$A = \pi r^2$$

Area:

453.6 yd

$$3.14 \cdot 12^2$$

Circumference:

37.6

a

yd²

b

yds

c

yd²

d

yds